

GROUP 1 – THE ALKALI METALS

KEY VOCABULARY

Alkali metal: a Group 1 metal with one outer electron.

Hydroxide: alkaline solution formed when Group 1 reacts with water.

Reactivity: increases going down Group 1.

Ion formed: Group 1 atoms lose 1 electron to form a 1+ ion.

COMMON MISCONCEPTIONS

~~Reactivity decreases down Group 1.~~

✓ Reactivity increases down the group.

~~Alkali metals are hard and dense.~~

✓ They are soft and have low density.

~~They form 2+ ions.~~

✓ They lose one electron to form 1+ ions.

REACTIVITY TREND

Lithium (Li)

Sodium (Na)

Potassium (K)



1 OUTER ELECTRONS

State the number of electrons in the **outer shell** of a Group 1 atom. [1 mark]

2 REACTION WITH WATER

Name the two products when sodium reacts with water. [2 marks]

3 EXPLAIN THE TREND

Explain why reactivity increases down Group 1. [3 marks]

Think about the distance of the outer electron from the nucleus.

4 BALANCE THE EQUATION

Complete: $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{_____} + \text{H}_2$. [1 mark]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

A Group 1 atom _____ one electron. Reactivity _____ down the group.

loses

increases

gains

decreases

6 PREDICT

Predict how potassium reacts with water compared with lithium. [2 marks]